Module 7

Capital Budgeting

FINE 3010, Financial Management

Prof. Renping Li, Tulane University

Goals

- How to decide what project(s) to do? The tool is the NPV rule

Goals

- How to decide what project(s) to do? The tool is the NPV rule
- But to use NPV, we need a project's cash flows

Goals

- How to decide what project(s) to do? The tool is the NPV rule
- But to use NPV, we need a project's cash flows
- Here, combine what we know about accounting and cash flow effects (**Module 2**) with the NPV rule (**Module 6**), to decide whether or not to accept a project

Goals

- How to decide what project(s) to do? The tool is the NPV rule
- But to use NPV, we need a project's cash flows
- Here, combine what we know about accounting and cash flow effects (**Module 2**) with the NPV rule (**Module 6**), to decide whether or not to accept a project
- **Capital budgeting**, the process of analyzing capital investment opportunities ("projects") and deciding which ones to accept

Roadmap

Should we invest in a project (e.g., a data center)?

NPV > 0?

Recall from **Module 6**

Roadmap

Should we invest in a project (e.g., a data center)?



Recall from **Module 6**

Roadmap

Should we invest in a project (e.g., a data center)?

How to forecast cash flows?

- Similar to **Module 2**, indirect method for cash flow
- But different, cash flow from a project, not a whole firm

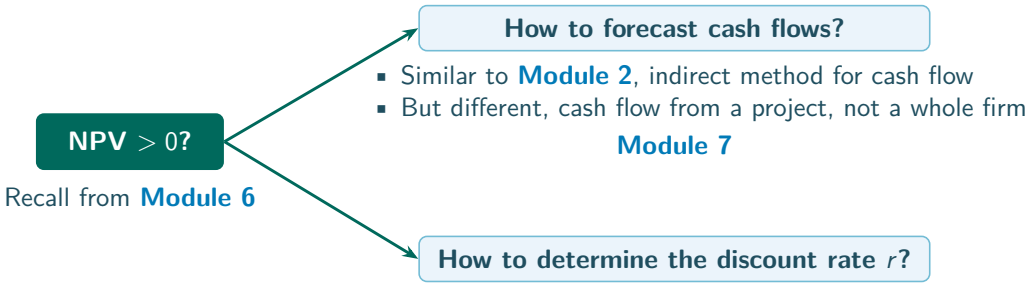
Module 7

NPV > 0?

Recall from **Module 6**

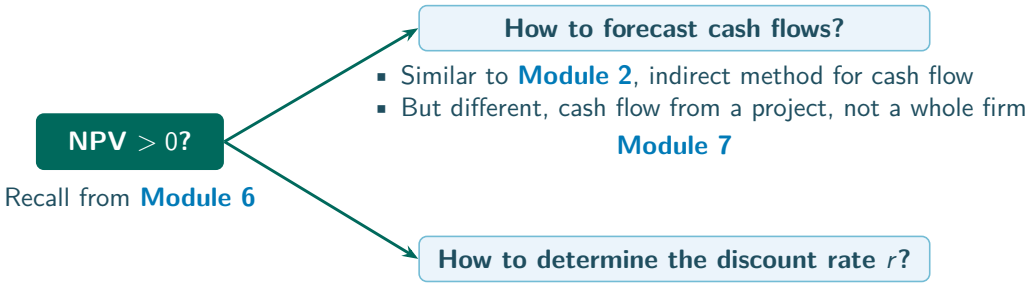
Roadmap

Should we invest in a project (e.g., a data center)?



Roadmap

Should we invest in a project (e.g., a data center)?



Motivating Example, Is AI Spending a Bubble?



CNBC, August 12, 2026 (7:21), "We're 'absolutely' in an AI bubble right now, says Lead Edge Capital's Mitchell Green"

Module 7, Part 0

Review of the Statement of Cash Flows

Statement of Cash Flows

Recall from **Module 2**

- **Cash flow**, cash actually received minus cash actually paid, no matter when the sale was made

Statement of Cash Flows

Recall from **Module 2**

- **Cash flow**, cash actually received minus cash actually paid, no matter when the sale was made
- The **statement of cash flows**, cash receipts and payments over a period

Statement of Cash Flows

Recall from **Module 2**

- **Cash flow**, cash actually received minus cash actually paid, no matter when the sale was made
- The **statement of cash flows**, cash receipts and payments over a period
- The statement has three parts
  - **Operating activities**, cash from running the business day to day

Statement of Cash Flows

Recall from **Module 2**

- **Cash flow**, cash actually received minus cash actually paid, no matter when the sale was made
- The **statement of cash flows**, cash receipts and payments over a period
- The statement has three parts
  - **Operating activities**, cash from running the business day to day
  - **Investing activities**, cash spent buying fixed assets and cash received from selling them

Statement of Cash Flows

Recall from **Module 2**

- **Cash flow**, cash actually received minus cash actually paid, no matter when the sale was made
- The **statement of cash flows**, cash receipts and payments over a period
- The statement has three parts
  - **Operating activities**, cash from running the business day to day
  - **Investing activities**, cash spent buying fixed assets and cash received from selling them
  - **Financing activities**, cash raised from lenders and shareholders, and cash returned to them

Statement of Cash Flows

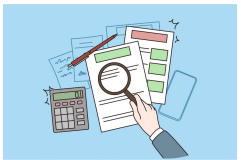
Recall from **Module 2**

Direct method	Indirect method
Lists every cash receipt and payment	Starts from net income and adjusts it
More time-consuming to prepare	Faster to prepare
Rarely used in practice	**The common choice**

Direct →



Indirect →



Statement of Cash Flows

Recall from **Module 2**

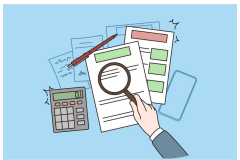
Direct method	Indirect method
Lists every cash receipt and payment	Starts from net income and adjusts it
More time-consuming to prepare	Faster to prepare
Rarely used in practice	**The common choice**

- Both methods give the same totals

Direct →



Indirect →



Statement of Cash Flows

Recall from **Module 2**

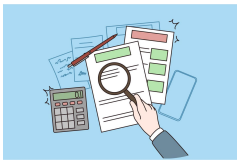
Direct method	Indirect method
Lists every cash receipt and payment	Starts from net income and adjusts it
More time-consuming to prepare	Faster to prepare
Rarely used in practice	**The common choice**

- Both methods give the same totals
- We will follow the indirect method, starting from net income in the income statement

Direct →



Indirect →



How Do Net Income and Cash Flow Differ?

Recall from **Module 2**

- The firm pays $2 billion to build a data center that lasts 5 years

How Do Net Income and Cash Flow Differ?

Recall from **Module 2**

- The firm pays $2 billion to build a data center that lasts 5 years
- We spread the cost over these 5 years

How Do Net Income and Cash Flow Differ?

Recall from **Module 2**

- The firm pays $2 billion to build a data center that lasts 5 years
- We spread the cost over these 5 years
- Each year, the depreciation is $0.4 billion

How Do Net Income and Cash Flow Differ?

Recall from **Module 2**

- The firm pays \$2 billion to build a data center that lasts 5 years
- We spread the cost over these 5 years
- Each year, the depreciation is \$0.4 billion

Year	0	1	2	3	4	5	
Cash flow (\$ billions)	**−2.0**	0	0	0	0	0	
Effect on net income	0	**−0.4**	**−0.4**	**−0.4**	**−0.4**	**−0.4**	← Depreciation

How Do Net Income and Cash Flow Differ?

Recall from **Module 2**

- The firm pays \$2 billion to build a data center that lasts 5 years
- We spread the cost over these 5 years
- Each year, the depreciation is \$0.4 billion

Year	0	1	2	3	4	5	
Cash flow (\$ billions)	**−2.0**	0	0	0	0	0	
Effect on net income	0	**−0.4**	**−0.4**	**−0.4**	**−0.4**	**−0.4**	← Depreciation

- Cash leaves all at once, net income feels it slowly

How Do Net Income and Cash Flow Differ?

Recall from **Module 2**

- The firm pays \$2 billion to build a data center that lasts 5 years
- We spread the cost over these 5 years
- Each year, the depreciation is \$0.4 billion

Year	0	1	2	3	4	5	
Cash flow (\$ billions)	**−2.0**	0	0	0	0	0	
Effect on net income	0	**−0.4**	**−0.4**	**−0.4**	**−0.4**	**−0.4**	← Depreciation

- Cash leaves all at once, net income feels it slowly

Relationship of cash flow (CF) and net income (NI)?

How Do Net Income and Cash Flow Differ?

Recall from **Module 2**

- The firm pays \$2 billion to build a data center that lasts 5 years
- We spread the cost over these 5 years
- Each year, the depreciation is \$0.4 billion

Year	0	1	2	3	4	5	
Cash flow (\$ billions)	**−2.0**	0	0	0	0	0	
Effect on net income	0	**−0.4**	**−0.4**	**−0.4**	**−0.4**	**−0.4**	← Depreciation

- Cash leaves all at once, net income feels it slowly

Relationship of cash flow (CF) and net income (NI)?
When spending on fixed assets, CF = NI − spending on fixed assets

How Do Net Income and Cash Flow Differ?

Recall from **Module 2**

- The firm pays \$2 billion to build a data center that lasts 5 years
- We spread the cost over these 5 years
- Each year, the depreciation is \$0.4 billion

Year	0	1	2	3	4	5	
Cash flow (\$ billions)	**−2.0**	0	0	0	0	0	
Effect on net income	0	**−0.4**	**−0.4**	**−0.4**	**−0.4**	**−0.4**	← Depreciation

- Cash leaves all at once, net income feels it slowly

Relationship of cash flow (CF) and net income (NI)?
When spending on fixed assets, CF = NI − spending on fixed assets
When recording depreciation, CF = NI + depreciation

How Do Net Income and Cash Flow Differ?

Recall from **Module 2**

- The firm spends $60 million producing AI control panels
- None of them have been sold yet



OpenAI, July 15, 2026 (2:15),
"Introducing the Codex Micro"

How Do Net Income and Cash Flow Differ?

Recall from **Module 2**

- The firm spends \$60 million producing AI control panels
- None of them have been sold yet

	Effect
Cash flow (\$ millions)	**−60**
Effect on net income	**0**
Change in inventories	**+60**



OpenAI, July 15, 2026 (2:15),
"Introducing the Codex Micro"

How Do Net Income and Cash Flow Differ?

Recall from **Module 2**

- The firm spends \$60 million producing AI control panels
- None of them have been sold yet

	Effect
Cash flow (\$ millions)	**−60**
Effect on net income	**0**
Change in inventories	**+60**

- The income statement records revenues and cost of goods sold (COGS) at the time of sale



OpenAI, July 15, 2026 (2:15), "Introducing the Codex Micro"

How Do Net Income and Cash Flow Differ?

Recall from **Module 2**

- The firm spends \$60 million producing AI control panels
- None of them have been sold yet

	Effect
Cash flow (\$ millions)	**−60**
Effect on net income	**0**
Change in inventories	**+60**

- The income statement records revenues and cost of goods sold (COGS) at the time of sale
- COGS is subtracted only when the panels are sold, not when they are produced



OpenAI, July 15, 2026 (2:15),
"Introducing the Codex Micro"

How Do Net Income and Cash Flow Differ?

Recall from **Module 2**

- The firm spends \$60 million producing AI control panels
- None of them have been sold yet

	Effect
Cash flow (\$ millions)	**−60**
Effect on net income	**0**
Change in inventories	**+60**

- The income statement records revenues and cost of goods sold (COGS) at the time of sale
- COGS is subtracted only when the panels are sold, not when they are produced



OpenAI, July 15, 2026 (2:15),
"Introducing the Codex Micro"

When adding to inventories, CF = NI − change in inventories

How Do Net Income and Cash Flow Differ?

Recall from **Module 2**

- The firm sells AI control panels for $100 million
- The customer has not paid yet



OpenAI, July 15, 2026 (2:15),
"Introducing the Codex Micro"

How Do Net Income and Cash Flow Differ?

Recall from **Module 2**

- The firm sells AI control panels for $100 million
- The customer has not paid yet

	Effect
Cash flow ($ millions)	**0**
Effect on net income	**+100**
Change in accounts receivable	**+100**



OpenAI, July 15, 2026 (2:15),
"Introducing the Codex Micro"

How Do Net Income and Cash Flow Differ?

Recall from **Module 2**

- The firm sells AI control panels for $100 million
- The customer has not paid yet

	Effect
Cash flow ($ millions)	**0**
Effect on net income	**+100**
Change in accounts receivable	**+100**

- Net income rises now, cash flow comes later when customers actually pay



OpenAI, July 15, 2026 (2:15), "Introducing the Codex Micro"

How Do Net Income and Cash Flow Differ?

Recall from **Module 2**

- The firm sells AI control panels for $100 million
- The customer has not paid yet

	Effect
Cash flow ($ millions)	**0**
Effect on net income	**+100**
Change in accounts receivable	**+100**

- Net income rises now, cash flow comes later when customers actually pay



OpenAI, July 15, 2026 (2:15), "Introducing the Codex Micro"

When firm not yet paid by customers, CF = NI − change in accounts receivable

How Do Net Income and Cash Flow Differ?

Recall from **Module 2**

- The firm owed suppliers $30 million for electronic parts
- The firm pays down the $30 million



OpenAI, July 15, 2026 (2:15),
"Introducing the Codex Micro"

How Do Net Income and Cash Flow Differ?

Recall from **Module 2**

- The firm owed suppliers \$30 million for electronic parts
- The firm pays down the \$30 million

	Effect
Cash flow (\$ millions)	**−30**
Effect on net income	**0**
Change in accounts payable	**−30**



OpenAI, July 15, 2026 (2:15),
"Introducing the Codex Micro"

How Do Net Income and Cash Flow Differ?

Recall from **Module 2**

- The firm owed suppliers \$30 million for electronic parts
- The firm pays down the \$30 million

	Effect
Cash flow (\$ millions)	**−30**
Effect on net income	**0**
Change in accounts payable	**−30**

- No sale of panels happened. No revenues or COGS should be recorded. Net income is 0



OpenAI, July 15, 2026 (2:15),
"Introducing the Codex Micro"

How Do Net Income and Cash Flow Differ?

Recall from **Module 2**

- The firm owed suppliers \$30 million for electronic parts
- The firm pays down the \$30 million

	Effect
Cash flow (\$ millions)	**−30**
Effect on net income	**0**
Change in accounts payable	**−30**

- No sale of panels happened. No revenues or COGS should be recorded. Net income is 0



OpenAI, July 15, 2026 (2:15),
"Introducing the Codex Micro"

How Do Net Income and Cash Flow Differ?

Recall from **Module 2**

- The firm owed suppliers \$30 million for electronic parts
- The firm pays down the \$30 million

	Effect
Cash flow (\$ millions)	**−30**
Effect on net income	**0**
Change in accounts payable	**−30**

- No sale of panels happened. No revenues or COGS should be recorded. Net income is 0



OpenAI, July 15, 2026 (2:15), "Introducing the Codex Micro"

When firm pays off suppliers, CF = NI + change in accounts payable

Operating Activities

Recall from **Module 2**

Operating activities, cash from running the business day to day

Steps to calculate cash flow from operating activities,

1. **Start** with net income
2. **Add** depreciation
3. **Subtract** change in net working capital (NWC)
   - NWC = inventories + accounts receivable − accounts payable
   - NWC, cash tied up in day-to-day operations, in unsold goods and unpaid bills

Why Subtract Change in Net Working Capital (NWC)?

Recall from **Module 2**

When adding to inventories, CF = NI − change in inventories

When firm not yet paid by customers, CF = NI − change in accounts receivable

When firm pays off suppliers, CF = NI + change in accounts payable

Why Subtract Change in Net Working Capital (NWC)?

Recall from **Module 2**

When adding to inventories, CF = NI − change in inventories

When firm not yet paid by customers, CF = NI − change in accounts receivable

When firm pays off suppliers, CF = NI + change in accounts payable

Ignoring other adjustments,

Why Subtract Change in Net Working Capital (NWC)?

Recall from **Module 2**

When adding to inventories, CF = NI − change in inventories

When firm not yet paid by customers, CF = NI − change in accounts receivable

When firm pays off suppliers, CF = NI + change in accounts payable

Ignoring other adjustments,

$$\begin{aligned} CF = NI \\ &- \text{change in inventories} \\ &- \text{change in accounts receivable} \\ &+ \text{change in accounts payable} \end{aligned}$$

Why Subtract Change in Net Working Capital (NWC)?

Recall from **Module 2**

When adding to inventories, CF = NI − change in inventories

When firm not yet paid by customers, CF = NI − change in accounts receivable

When firm pays off suppliers, CF = NI + change in accounts payable

Ignoring other adjustments,

$$\begin{aligned} CF = NI \\ &- \text{change in inventories} \\ &- \text{change in accounts receivable} \\ &+ \text{change in accounts payable} \end{aligned}$$

$$CF = NI - \text{change in (inventories} + \text{accounts receivable} - \text{accounts payable)}$$

Why Subtract Change in Net Working Capital (NWC)?

Recall from **Module 2**

When adding to inventories, CF = NI − change in inventories

When firm not yet paid by customers, CF = NI − change in accounts receivable

When firm pays off suppliers, CF = NI + change in accounts payable

Ignoring other adjustments,

$$\begin{aligned} CF = NI \\ &- \text{change in inventories} \\ &- \text{change in accounts receivable} \\ &+ \text{change in accounts payable} \end{aligned}$$

$$CF = NI - \text{change in (inventories} + \text{accounts receivable} - \text{accounts payable)}$$

$$CF = NI - \text{change in net working capital (NWC)}$$

Investing Activities

Recall from **Module 2**

Investing activities, cash spent buying fixed assets and cash received from selling them

Investing activities show cash flows from,

1. Capital purchased and sold
   - **Capital**, a firm's fixed assets
   - **Capital expenditure**, the amount spent on buying capital during the period
   - Cash flow is *negative* if firms buy capital

Financing Activities

Recall from **Module 2**

Financing activities, cash raised from lenders and shareholders, and cash returned to them

Recap of Part 0

Recall from **Module 2**

- The statement of cash flows, cash receipts and payments over a period
- Net income $\neq$ cash flow
- The statement has three parts
  - **Operating activities**, cash flow = Net income + Depreciation − Change in NWC
  - **Investing activities**, cash spent buying fixed assets and cash received from selling them
  - **Financing activities**, cash raised from lenders and shareholders, and cash returned to them

Module 7, Part 1

Project Cash Flow

Firm Cash Flow vs. Project Cash Flow

- **Module 2** looked at the whole firm, here we look at one project on its own

Firm Cash Flow vs. Project Cash Flow

- **Module 2** looked at the whole firm, here we look at one project on its own
- **Operating activities**, the same idea, project net income plus depreciation minus the change in NWC

Firm Cash Flow vs. Project Cash Flow

- **Module 2** looked at the whole firm, here we look at one project on its own
- **Operating activities**, the same idea, project net income plus depreciation minus the change in NWC
- **Investing activities**, the project's capital investments, buying its assets at the start

Firm Cash Flow vs. Project Cash Flow

- **Module 2** looked at the whole firm, here we look at one project on its own
- **Operating activities**, the same idea, project net income plus depreciation minus the change in NWC
- **Investing activities**, the project's capital investments, buying its assets at the start
- **Financing activities**, *none*, we do not consider financing activities when evaluating a project on its own

Identify Project Cash Flow

- Cash flow from a project has four components,

Identify Project Cash Flow

- Cash flow from a project has four components,

$$\text{Cash flow from a project} = \begin{cases} \text{Project net income} & \text{Component 1} \\ + \text{ Depreciation} & \text{Component 2} \\ \text{Cash flow from changes in NWC} & \text{Component 3} \\ \text{Cash flow from capital investments} & \text{Component 4} \end{cases}$$

Component 1, Project Net Income

- While the project operates, it earns revenues and incurs expenses

Component 1, Project Net Income

- While the project operates, it earns revenues and incurs expenses
- Build the project's own income statement

Component 1, Project Net Income

- While the project operates, it earns revenues and incurs expenses
- Build the project's own income statement

Revenues from the project
– COGS and SG&A expenses for the project
– Depreciation for the project's capital investment
Project EBIT
– Taxes (tax rate $\times$ project EBIT)
Project net income

Component 2, Depreciation

- Depreciation is subtracted on the project's income statement

Component 2, Depreciation

- Depreciation is subtracted on the project's income statement
- It is not a cash flow! Instead, the cash went out at the time of the project's capital investment

Component 2, Depreciation

- Depreciation is subtracted on the project's income statement
- It is not a cash flow! Instead, the cash went out at the time of the project's capital investment
- To get cash flow from the project, add back depreciation, as in **Module 2**

Example, Project Net Income and Depreciation

- A project requires an investment in fixed assets of $2,000. The project's revenues minus its COGS and SG&A expenses are $1,100 in year 1 and $1,300 in year 2. The investment is depreciated at $1,000 per year. The corporate tax rate is 21%

Example, Project Net Income and Depreciation

- A project requires an investment in fixed assets of $2,000. The project's revenues minus its COGS and SG&A expenses are $1,100 in year 1 and $1,300 in year 2. The investment is depreciated at $1,000 per year. The corporate tax rate is 21%
- Calculate the project's net income plus depreciation for year 1

Example, Project Net Income and Depreciation

- Solution,

Item	$
Revenues − COGS − SG&A expenses	1,100
− Depreciation	1,000
Project EBIT	**100**
− Taxes	21
Project net income	**79**
+ Depreciation	1,000
Project net income + Depreciation	**1,079**

Example, Project Net Income and Depreciation

- Solution,

Item	$
Revenues − COGS − SG&A expenses	1,100
− Depreciation	1,000
Project EBIT	**100**
− Taxes	21
Project net income	**79**
+ Depreciation	1,000
Project net income + Depreciation	**1,079**

- Taxes, $21\% \times$ project EBIT $= 21\% \times 100 = 21$

Example, Project Net Income and Depreciation

- Solution,

Item	$
Revenues − COGS − SG&A expenses	1,100
− Depreciation	1,000
Project EBIT	**100**
− Taxes	21
Project net income	**79**
+ Depreciation	1,000
Project net income + Depreciation	**1,079**

- Taxes, $21\% \times$ project EBIT $= 21\% \times 100 = 21$
- Project net income + Depreciation $= 79 + 1{,}000 = \$1{,}079$

Component 3, Cash Flow from Changes in NWC

- Recall from **Module 2**,
  - NWC = inventories + accounts receivable − accounts payable
  - It is cash tied up in day-to-day operations, in unsold goods and unpaid bills

Component 3, Cash Flow from Changes in NWC

- Recall from **Module 2**,
  - NWC = inventories + accounts receivable − accounts payable
  - It is cash tied up in day-to-day operations, in unsold goods and unpaid bills
- A project has its own NWC, e.g.,
  - an inventory of raw materials before production starts
  - accounts receivable rising as sales start
  - accounts payable rising as the project buys supplies on credit

Component 3, Cash Flow from Changes in NWC

- Recall from **Module 2**,
  - NWC = inventories + accounts receivable − accounts payable
  - It is cash tied up in day-to-day operations, in unsold goods and unpaid bills
- A project has its own NWC, e.g.,
  - an inventory of raw materials before production starts
  - accounts receivable rising as sales start
  - accounts payable rising as the project buys supplies on credit
- An increase in the project's NWC is a negative cash flow, and a decrease is a positive cash flow,

Cash flow from changes in NWC = −Change in the project's NWC

Component 3, Cash Flow from Changes in NWC

- E.g., a 3-year project needs $100 of NWC while it operates, recovered when the project ends

Component 3, Cash Flow from Changes in NWC

- E.g., a 3-year project needs \$100 of NWC while it operates, recovered when the project ends

Year	0	1	2	3
The project's NWC	100	100	100	0
Change in NWC	$+100$	0	0	-100
Cash flow from changes in NWC	-100	0	0	$+100$

Component 3, Cash Flow from Changes in NWC

- E.g., a 3-year project needs \$100 of NWC while it operates, recovered when the project ends

Year	0	1	2	3
The project's NWC	100	100	100	0
Change in NWC	$+100$	0	0	-100
Cash flow from changes in NWC	-100	0	0	$+100$

- The \$100 goes out at $t = 0$ and returns only at $t = 3$, so tying up the cash still costs the time value of money

Component 4, Cash Flow from Capital Investments

- If the project requires its own investment in fixed assets to get started (e.g., building a data center), that is a negative cash flow

Component 4, Cash Flow from Capital Investments

- If the project requires its own investment in fixed assets to get started (e.g., building a data center), that is a negative cash flow
- Fixed assets are usually depreciated "straight-line" over time,
  - **Expected life**, the number of years the project uses the fixed assets

$$\text{Annual depreciation} = \frac{\text{Capital investment}}{\text{Expected life}}$$

Identify Project Cash Flow

- Cash flow from a project has four components,

$$\text{Cash flow from a project} = \begin{cases} \text{Project net income} & \text{Component 1} \\ + \text{ Depreciation} & \text{Component 2} \\ \text{Cash flow from changes in NWC} & \text{Component 3} \\ \text{Cash flow from capital investments} & \text{Component 4} \end{cases}$$

Recap of Part 1

- Project net income comes from the project's own income statement, revenues less COGS and SG&A expenses, depreciation, and taxes

Recap of Part 1

- Project net income comes from the project's own income statement, revenues less COGS and SG&A expenses, depreciation, and taxes
- Depreciation is not a cash flow, add it back to the project's net income

Recap of Part 1

- Project net income comes from the project's own income statement, revenues less COGS and SG&A expenses, depreciation, and taxes
- Depreciation is not a cash flow, add it back to the project's net income
- An increase in the project's NWC is a negative cash flow, and a decrease is a positive cash flow

Recap of Part 1

- Project net income comes from the project's own income statement, revenues less COGS and SG&A expenses, depreciation, and taxes
- Depreciation is not a cash flow, add it back to the project's net income
- An increase in the project's NWC is a negative cash flow, and a decrease is a positive cash flow
- The investment in fixed assets is a negative cash flow, and the fixed assets are depreciated straight-line over the expected life

Module 7, Part 2

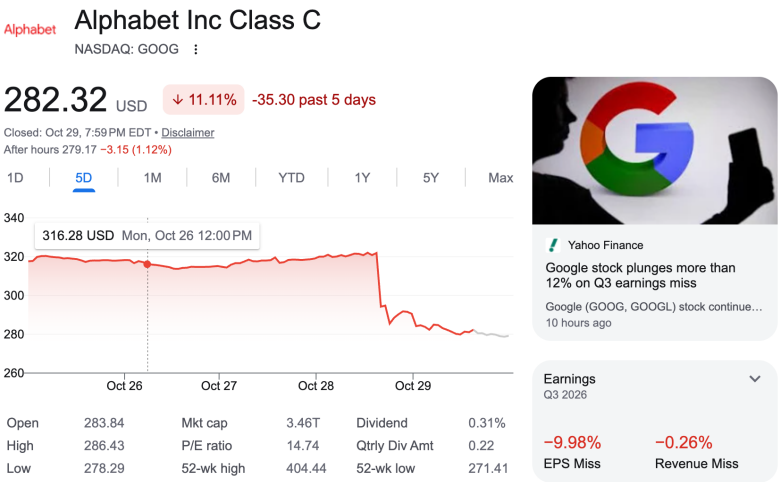
Putting It All Together

Motivating Example, Google's AI Spend under Scrutiny



CNBC, October 28, 2026 (3:06), "Google's AI spend under scrutiny, here's what to know"

Motivating Example, Google's AI Spend under Scrutiny



Example, Putting It All Together

- A company is considering building a new data center, an investment in fixed assets of $1 billion ($1,000 million) that will last 5 years. It will be depreciated straight-line over 5 years

Example, Putting It All Together

- A company is considering building a new data center, an investment in fixed assets of \$1 billion (\$1,000 million) that will last 5 years. It will be depreciated straight-line over 5 years
- The data center needs an increase in NWC of \$100 million at $t = 0$ (spare servers and parts), recovered fully at $t = 5$

Example, Putting It All Together

- A company is considering building a new data center, an investment in fixed assets of \$1 billion (\$1,000 million) that will last 5 years. It will be depreciated straight-line over 5 years
- The data center needs an increase in NWC of \$100 million at $t = 0$ (spare servers and parts), recovered fully at $t = 5$
- Renting out its computing capacity brings in \$240 million a year after COGS and SG&A expenses (electricity, staff, and so on)

Example, Putting It All Together

- A company is considering building a new data center, an investment in fixed assets of \$1 billion (\$1,000 million) that will last 5 years. It will be depreciated straight-line over 5 years
- The data center needs an increase in NWC of \$100 million at $t = 0$ (spare servers and parts), recovered fully at $t = 5$
- Renting out its computing capacity brings in \$240 million a year after COGS and SG&A expenses (electricity, staff, and so on)
- The corporate tax rate is 21%, and the discount rate is 8%

Example, Putting It All Together

- A company is considering building a new data center, an investment in fixed assets of \$1 billion (\$1,000 million) that will last 5 years. It will be depreciated straight-line over 5 years
- The data center needs an increase in NWC of \$100 million at $t = 0$ (spare servers and parts), recovered fully at $t = 5$
- Renting out its computing capacity brings in \$240 million a year after COGS and SG&A expenses (electricity, staff, and so on)
- The corporate tax rate is 21%, and the discount rate is 8%
- Should the company build the data center or not?

Identify Project Cash Flow

- Cash flow from a project has four components,

$$\text{Cash flow from a project} = \begin{cases} \text{Project net income} & \text{Component 1} \\ + \text{ Depreciation} & \text{Component 2} \\ \text{Cash flow from changes in NWC} & \text{Component 3} \\ \text{Cash flow from capital investments} & \text{Component 4} \end{cases}$$

Example, Component 1, Project Net Income

- Build the project's income statement, the same in years 1 to 5 ($ millions),

Year	0	1	2	3	4	5

Example, Component 1, Project Net Income

- Build the project's income statement, the same in years 1 to 5 ($ millions),

Year	0	1	2	3	4	5
Revenues − COGS − SG&A expenses		240	240	240	240	240

Example, Component 1, Project Net Income

- Build the project's income statement, the same in years 1 to 5 ($ millions),

Year	0	1	2	3	4	5
Revenues − COGS − SG&A expenses		240	240	240	240	240
− Depreciation		200	200	200	200	200

- Depreciation $= 1{,}000/5 = 200$ a year

Example, Component 1, Project Net Income

- Build the project's income statement, the same in years 1 to 5 ($ millions),

Year	0	1	2	3	4	5
Revenues − COGS − SG&A expenses		240	240	240	240	240
− Depreciation		200	200	200	200	200
Project EBIT		**40**	**40**	**40**	**40**	**40**

- Depreciation $= 1{,}000/5 = 200$ a year

Example, Component 1, Project Net Income

- Build the project's income statement, the same in years 1 to 5 ($ millions),

Year	0	1	2	3	4	5
Revenues − COGS − SG&A expenses		240	240	240	240	240
− Depreciation		200	200	200	200	200
Project EBIT		**40**	**40**	**40**	**40**	**40**
− Taxes		8.4	8.4	8.4	8.4	8.4

- Depreciation $= 1{,}000/5 = 200$ a year
- Taxes $= 21\% \times 40 = 8.4$

Example, Component 1, Project Net Income

- Build the project's income statement, the same in years 1 to 5 ($ millions),

Year	0	1	2	3	4	5
Revenues − COGS − SG&A expenses		240	240	240	240	240
− Depreciation		200	200	200	200	200
Project EBIT		**40**	**40**	**40**	**40**	**40**
− Taxes		8.4	8.4	8.4	8.4	8.4
Project net income		**31.6**	**31.6**	**31.6**	**31.6**	**31.6**

- Depreciation $= 1{,}000/5 = 200$ a year
- Taxes $= 21\% \times 40 = 8.4$

Example, Component 2, Depreciation

- Depreciation is not a cash flow, add it back ($ millions),

Year	0	1	2	3	4	5

Example, Component 2, Depreciation

- Depreciation is not a cash flow, add it back ($ millions),

Year	0	1	2	3	4	5
Project net income		31.6	31.6	31.6	31.6	31.6

Example, Component 2, Depreciation

- Depreciation is not a cash flow, add it back ($ millions),

Year	0	1	2	3	4	5
Project net income		31.6	31.6	31.6	31.6	31.6
+ Depreciation		200	200	200	200	200

Example, Component 2, Depreciation

- Depreciation is not a cash flow, add it back ($ millions),

Year	0	1	2	3	4	5
Project net income		31.6	31.6	31.6	31.6	31.6
+ Depreciation		200	200	200	200	200
Project net income + Depreciation		**231.6**	**231.6**	**231.6**	**231.6**	**231.6**

- The cash for the data center went out at $t = 0$, it is counted in Component 4

Example, Component 3, Cash Flow from Changes in NWC

- The project's NWC is 100 while it operates, recovered at $t = 5$ ($ millions),

Year	0	1	2	3	4	5

Example, Component 3, Cash Flow from Changes in NWC

- The project's NWC is 100 while it operates, recovered at $t = 5$ ($ millions),

Year	0	1	2	3	4	5
The project's NWC	100	100	100	100	100	0

Example, Component 3, Cash Flow from Changes in NWC

- The project's NWC is 100 while it operates, recovered at $t = 5$ ($ millions),

Year	0	1	2	3	4	5
The project's NWC	100	100	100	100	100	0
Change in NWC	+100	0	0	0	0	−100

Example, Component 3, Cash Flow from Changes in NWC

- The project's NWC is 100 while it operates, recovered at $t = 5$ ($ millions),

Year	0	1	2	3	4	5
The project's NWC	100	100	100	100	100	0
Change in NWC	+100	0	0	0	0	−100
Cash flow from changes in NWC	**−100**	**0**	**0**	**0**	**0**	**+100**

- An increase in NWC is a negative cash flow, and a decrease is a positive cash flow

Example, Component 4, Cash Flow from Capital Investments

- Investment in fixed assets at $t = 0$ ($ millions),

Year	0	1	2	3	4	5

Example, Component 4, Cash Flow from Capital Investments

- Investment in fixed assets at $t = 0$ ($ millions),

Year	0	1	2	3	4	5
Investment in fixed assets	−1,000	0	0	0	0	0

Example, Component 4, Cash Flow from Capital Investments

- Investment in fixed assets at $t = 0$ ($ millions),

Year	0	1	2	3	4	5
Investment in fixed assets	−1,000	0	0	0	0	0
Cash flow from capital investments	**−1,000**	**0**	**0**	**0**	**0**	**0**

Example, Total Cash Flows

- Add up the four components year by year ($ millions),

Year	0	1	2	3	4	5

Example, Total Cash Flows

- Add up the four components year by year ($ millions),

Year	0	1	2	3	4	5
Project net income + Depreciation	0	231.6	231.6	231.6	231.6	231.6

Example, Total Cash Flows

- Add up the four components year by year ($ millions),

Year	0	1	2	3	4	5
Project net income + Depreciation	0	231.6	231.6	231.6	231.6	231.6
Cash flow from changes in NWC	−100	0	0	0	0	+100

Example, Total Cash Flows

- Add up the four components year by year ($ millions),

Year	0	1	2	3	4	5
Project net income + Depreciation	0	231.6	231.6	231.6	231.6	231.6
Cash flow from changes in NWC	−100	0	0	0	0	+100
Cash flow from capital investments	−1,000	0	0	0	0	0

Example, Total Cash Flows

- Add up the four components year by year ($ millions),

Year	0	1	2	3	4	5
Project net income + Depreciation	0	231.6	231.6	231.6	231.6	231.6
Cash flow from changes in NWC	−100	0	0	0	0	+100
Cash flow from capital investments	−1,000	0	0	0	0	0
Total cash flows	**−1,100**	**231.6**	**231.6**	**231.6**	**231.6**	**331.6**

- At $t = 0$, $-1{,}000 - 100 = -1{,}100$. At $t = 5$, $231.6 + 100 = 331.6$

Example, NPV and the Decision

- Discount the total cash flows at the 8% discount rate ($ millions),

Example, NPV and the Decision

- Discount the total cash flows at the 8% discount rate ($ millions),

$$\text{NPV} = -1{,}100 + \frac{231.6}{1.08} + \frac{231.6}{1.08^2} + \frac{231.6}{1.08^3} + \frac{231.6}{1.08^4} + \frac{331.6}{1.08^5} = -107.23$$

Example, NPV and the Decision

- Discount the total cash flows at the 8% discount rate ($ millions),

$$\text{NPV} = -1{,}100 + \frac{231.6}{1.08} + \frac{231.6}{1.08^2} + \frac{231.6}{1.08^3} + \frac{231.6}{1.08^4} + \frac{331.6}{1.08^5} = -107.23$$

- NPV < 0, so the company should not build the data center

Example, NPV with the Cash Flow Worksheet

One cash flow at a time

Enter	Keys
	CF, **2nd**, **CLR Work**
−1100	**ENTER**, ↓
231.6	**ENTER**, ↓ ↓
231.6	**ENTER**, ↓ ↓
231.6	**ENTER**, ↓ ↓
231.6	**ENTER**, ↓ ↓
331.6	**ENTER**
	NPV (the calculator asks for I)
8	**ENTER**, ↓
	CPT
−107.23	

Example, NPV with the Cash Flow Worksheet

One cash flow at a time

Enter	Keys
	CF, 2nd, CLR Work
−1100	ENTER, ↓
231.6	ENTER, ↓ ↓
231.6	ENTER, ↓ ↓
231.6	ENTER, ↓ ↓
231.6	ENTER, ↓ ↓
331.6	ENTER
	NPV (the calculator asks for I)
8	ENTER, ↓
	CPT
−107.23	

Repeated cash flow with frequency

Enter	Keys
	CF, 2nd, CLR Work
−1100	ENTER, ↓
231.6	ENTER, ↓
4	ENTER (the frequency F01), ↓
331.6	ENTER
	NPV (the calculator asks for I)
8	ENTER, ↓
	CPT
−107.23	

Example, IRR with the Cash Flow Worksheet

- Recall from **Module 6**, the IRR is the discount rate at which NPV $= 0$

Example, IRR with the Cash Flow Worksheet

- Recall from **Module 6**, the IRR is the discount rate at which $\text{NPV} = 0$
- The same cash flow worksheet, then **IRR**, **CPT**

Enter	Keys
	CF, **2nd**, **CLR Work**
-1100	**ENTER**, ↓
231.6	**ENTER**, ↓
4	**ENTER** (the frequency F01), ↓
331.6	**ENTER**
	IRR (shows 0)
	CPT
4.41%	

Example, IRR with the Cash Flow Worksheet

- Recall from **Module 6**, the IRR is the discount rate at which NPV $= 0$
- The same cash flow worksheet, then **IRR**, **CPT**
- IRR (4.41%) $<$ discount rate (8%), reject the project, the same answer as the NPV rule

Enter	Keys
	CF, **2nd**, **CLR Work**
-1100	**ENTER**, ↓
231.6	**ENTER**, ↓
4	**ENTER** (the frequency F01), ↓
331.6	**ENTER**
	IRR (shows 0)
	CPT
4.41%	

How to Attack a Full-Scale Problem

1. Build the project's income statement year by year, then add back depreciation, project net income + depreciation (Components 1 and 2)

How to Attack a Full-Scale Problem

1. Build the project's income statement year by year, then add back depreciation, project net income + depreciation (Components 1 and 2)
2. Lay out the project's NWC each year, then Cash flow from changes in NWC $= -$Change in the project's NWC (Component 3)

How to Attack a Full-Scale Problem

1. Build the project's income statement year by year, then add back depreciation, project net income + depreciation (Components 1 and 2)
2. Lay out the project's NWC each year, then Cash flow from changes in NWC $= -$Change in the project's NWC (Component 3)
3. Add the cash flow from capital investments, the investment in fixed assets at the start and any terminal cash flows at the end (Component 4)

How to Attack a Full-Scale Problem

1. Build the project's income statement year by year, then add back depreciation, project net income + depreciation (Components 1 and 2)
2. Lay out the project's NWC each year, then Cash flow from changes in NWC $= -$Change in the project's NWC (Component 3)
3. Add the cash flow from capital investments, the investment in fixed assets at the start and any terminal cash flows at the end (Component 4)
4. Total the cash flows year by year, discount at the discount rate, and apply the NPV rule

Module 7, Part 3

The Principles behind Project Cash Flows

General Principles for Identifying Project Cash Flow

- Look back at the data center table. It quietly followed three principles,

General Principles for Identifying Project Cash Flow

- Look back at the data center table. It quietly followed three principles,

1. Discount cash flows, not net income

General Principles for Identifying Project Cash Flow

- Look back at the data center table. It quietly followed three principles,

1. Discount cash flows, not net income
2. Exclude cash flows from financing activities

General Principles for Identifying Project Cash Flow

- Look back at the data center table. It quietly followed three principles,

1. Discount cash flows, not net income
2. Exclude cash flows from financing activities
3. Consider all and only incremental cash flows

Principle 1, Discount Cash Flows, Not Net Income

- Net income $\neq$ cash flows

Principle 1, Discount Cash Flows, Not Net Income

- Net income $\neq$ cash flows
- Differences in how capital investments, depreciation, and NWC are handled

Principle 1, Discount Cash Flows, Not Net Income

- Net income $\neq$ cash flows
- Differences in how capital investments, depreciation, and NWC are handled
- We always discount the actual cash flows to their present value

Principle 1, Discount Cash Flows, Not Net Income

- Net income $\neq$ cash flows
- Differences in how capital investments, depreciation, and NWC are handled
- We always discount the actual cash flows to their present value
- We do not discount net income

Example, Net Income vs. Cash Flows

- A project requires an investment in fixed assets of $2,000 and lasts 2 years. The fixed assets are depreciated straight-line at $1,000 per year

Example, Net Income vs. Cash Flows

- A project requires an investment in fixed assets of $2,000 and lasts 2 years. The fixed assets are depreciated straight-line at $1,000 per year
- The project's revenues minus its COGS and SG&A expenses are $1,500 in year 1 and $500 in year 2. There are no taxes and no NWC is required for this project

Example, Net Income vs. Cash Flows

- A project requires an investment in fixed assets of $2,000 and lasts 2 years. The fixed assets are depreciated straight-line at $1,000 per year
- The project's revenues minus its COGS and SG&A expenses are $1,500 in year 1 and $500 in year 2. There are no taxes and no NWC is required for this project
- The discount rate is 10%

Example, Net Income vs. Cash Flows

- A project requires an investment in fixed assets of $2,000 and lasts 2 years. The fixed assets are depreciated straight-line at $1,000 per year
- The project's revenues minus its COGS and SG&A expenses are $1,500 in year 1 and $500 in year 2. There are no taxes and no NWC is required for this project
- The discount rate is 10%
- Compare the NPV using net income to the NPV using cash flows

Example, Net Income vs. Cash Flows

- A project requires an investment in fixed assets of $2,000 and lasts 2 years. The fixed assets are depreciated straight-line at $1,000 per year
- The project's revenues minus its COGS and SG&A expenses are $1,500 in year 1 and $500 in year 2. There are no taxes and no NWC is required for this project
- The discount rate is 10%
- Compare the NPV using net income to the NPV using cash flows

Clearly a money-loser!

Example, Net Income vs. Cash Flows

- Solution,

Year	0	1	2

Example, Net Income vs. Cash Flows

- Solution,

Year	0	1	2
Revenues − COGS − SG&A expenses		1,500	500
− Depreciation		1,000	1,000

Example, Net Income vs. Cash Flows

- Solution,

Year	0	1	2
Revenues − COGS − SG&A expenses		1,500	500
− Depreciation		1,000	1,000
Project net income		**500**	**−500**

Example, Net Income vs. Cash Flows

- Solution,

Year	0	1	2
Revenues − COGS − SG&A expenses		1,500	500
− Depreciation		1,000	1,000
Project net income		**500**	**−500**
+ Depreciation		1,000	1,000
Project net income + Depreciation	**0**	**1,500**	**500**

Example, Net Income vs. Cash Flows

- Solution,

Year	0	1	2
Revenues − COGS − SG&A expenses		1,500	500
− Depreciation		1,000	1,000
Project net income		**500**	**−500**
+ Depreciation		1,000	1,000
Project net income + Depreciation	**0**	**1,500**	**500**
Cash flow from capital investments	−2,000	0	0

Example, Net Income vs. Cash Flows

- Solution,

Year	0	1	2
Revenues − COGS − SG&A expenses		1,500	500
− Depreciation		1,000	1,000
Project net income		**500**	**−500**
+ Depreciation		1,000	1,000
Project net income + Depreciation	**0**	**1,500**	**500**
Cash flow from capital investments	−2,000	0	0
Total cash flows	**−2,000**	**1,500**	**500**

Example, Net Income vs. Cash Flows

- Solution,

Year	0	1	2
Revenues − COGS − SG&A expenses		1,500	500
− Depreciation		1,000	1,000
Project net income		**500**	**−500**
+ Depreciation		1,000	1,000
Project net income + Depreciation	**0**	**1,500**	**500**
Cash flow from capital investments	−2,000	0	0
Total cash flows	**−2,000**	**1,500**	**500**

- NPV using net income $= \frac{500}{1.10} - \frac{500}{1.10^2} = \$41.32$, which says take the project

Example, Net Income vs. Cash Flows

- Solution,

Year	0	1	2
Revenues − COGS − SG&A expenses		1,500	500
− Depreciation		1,000	1,000
Project net income		**500**	**−500**
+ Depreciation		1,000	1,000
Project net income + Depreciation	**0**	**1,500**	**500**
Cash flow from capital investments	−2,000	0	0
Total cash flows	**−2,000**	**1,500**	**500**

- NPV using net income $= \frac{500}{1.10} - \frac{500}{1.10^2} = \$41.32$, which says take the project
- NPV using cash flows $= -2{,}000 + \frac{1{,}500}{1.10} + \frac{500}{1.10^2} = -\$223.14$

Example, Net Income vs. Cash Flows

- Solution,

Year	0	1	2
Revenues − COGS − SG&A expenses		1,500	500
− Depreciation		1,000	1,000
Project net income		**500**	**−500**
+ Depreciation		1,000	1,000
Project net income + Depreciation	**0**	**1,500**	**500**
Cash flow from capital investments	−2,000	0	0
Total cash flows	**−2,000**	**1,500**	**500**

- NPV using net income $= \frac{500}{1.10} - \frac{500}{1.10^2} = \$41.32$, which says take the project
- NPV using cash flows $= -2{,}000 + \frac{1{,}500}{1.10} + \frac{500}{1.10^2} = -\$223.14$
- NPV < 0, so the project should be rejected, net income gets it wrong

Principle 2, Exclude Cash Flows from Financing Activities

- Project cash flows exclude cash flows from financing activities

Principle 2, Exclude Cash Flows from Financing Activities

- Project cash flows exclude cash flows from financing activities
- The mini-income statement for this data center project has no interest expense row, even if the company borrows to build it

Principle 3, Consider All and Only Incremental Cash Flows

- We need to discount **incremental** cash flows
- What does "incremental" mean in capital budgeting?

Principle 3, Consider All and Only Incremental Cash Flows

- We need to discount **incremental** cash flows
- What does "incremental" mean in capital budgeting?
- Ask yourself, "would the cash flow still exist if the project did not exist?"
  - If yes, do *not* include it in the analysis
  - If no, include it

Principle 3, Consider All and Only Incremental Cash Flows

- We need to discount **incremental** cash flows
- What does "incremental" mean in capital budgeting?
- Ask yourself, "would the cash flow still exist if the project did not exist?"
  - If yes, do *not* include it in the analysis
  - If no, include it

Incremental cash flow = Cash flow with the project − Cash flow without it

Incremental Cash Flows, a Few Rules That Are Easy to Forget

a. Include all indirect effects

Incremental Cash Flows, a Few Rules That Are Easy to Forget

a. Include all indirect effects
b. Exclude sunk costs

Incremental Cash Flows, a Few Rules That Are Easy to Forget

a. Include all indirect effects
b. Exclude sunk costs
c. Include opportunity costs

Incremental Cash Flows, a Few Rules That Are Easy to Forget

a. Include all indirect effects
b. Exclude sunk costs
c. Include opportunity costs
d. Include cash flows from changes in NWC

Incremental Cash Flows, a Few Rules That Are Easy to Forget

a. Include all indirect effects
b. Exclude sunk costs
c. Include opportunity costs
d. Include cash flows from changes in NWC
e. Include terminal cash flows

Rule a, Indirect Effects

- **Indirect effects**, the effects of a project on the cash flows of the company's other businesses

Rule a, Indirect Effects

- **Indirect effects**, the effects of a project on the cash flows of the company's other businesses
- E.g., **cannibalization**, a new product cutting into the sales of the company's existing products

Rule a, Indirect Effects

- Nintendo plans a new-generation gaming console
- It requires an investment of $200m and produces cash flows of $400m and $800m in years 1 and 2





Rule a, Indirect Effects

- Nintendo plans a new-generation gaming console
- It requires an investment of $200m and produces cash flows of $400m and $800m in years 1 and 2
- Looks like a great investment!





Rule a, Indirect Effects

- Nintendo plans a new-generation gaming console
- It requires an investment of $200m and produces cash flows of $400m and $800m in years 1 and 2
- Looks like a great investment!
- But the launch will hurt sales of the older console
- Without the new console, the old one is forecast to create $800m and $1,000m in years 1 and 2, with the new console, only $500m and $400m





Rule a, Indirect Effects

- Nintendo plans a new-generation gaming console
- It requires an investment of $200m and produces cash flows of $400m and $800m in years 1 and 2
- Looks like a great investment!
- But the launch will hurt sales of the older console
- Without the new console, the old one is forecast to create $800m and $1,000m in years 1 and 2, with the new console, only $500m and $400m
- What is the **incremental** cash flow of the new console?





Rule a, Indirect Effects

- Solution ($ millions),

Year	0	1	2
Cash flow from investment in and sales of the new one	−200	400	800
Cash flow from the old console without the new one		800	1,000
Cash flow from the old console with the new one		500	400
Change in cash flow from the old console		−300	−600
Incremental cash flow of the new console	−200	$400 - 300 = 100$	$800 - 600 = 200$

Rule a, Indirect Effects

- Solution ($ millions),

Year	0	1	2	
Cash flow from investment in and sales of the new one	−200	400	800	← Direct effect
Cash flow from the old console without the new one		800	1,000	
Cash flow from the old console with the new one		500	400	
Change in cash flow from the old console		−300	−600	← Indirect effect
Incremental cash flow of the new console	−200	$400 - 300 = 100$	$800 - 600 = 200$	

Rule b, Sunk Costs

- **Sunk costs**, expenses that have already been paid, or expenses that can no longer be changed
  - E.g., previous R&D (already paid), rent under an unbreakable lease (can no longer be changed)



Rule b, Sunk Costs

- **Sunk costs**, expenses that have already been paid, or expenses that can no longer be changed
  - E.g., previous R&D (already paid), rent under an unbreakable lease (can no longer be changed)

- A sunk cost is the same with or without the project $\rightarrow$ it is not incremental and should not affect the NPV $\rightarrow$ Exclude it!



Rule b, Sunk Costs

- **Sunk costs**, expenses that have already been paid, or expenses that can no longer be changed
  - E.g., previous R&D (already paid), rent under an unbreakable lease (can no longer be changed)
- A sunk cost is the same with or without the project $\rightarrow$ it is not incremental and should not affect the NPV $\rightarrow$ Exclude it!
- Many sunk costs are "related" to the project. But if the decision does not change them, they are not incremental



Example, Sunk Costs

- Your company has developed a new product that is almost ready to launch. The R&D cost $0.6 million and was completed last year
- Launching requires another $1 million investment (marketing and so on), with a present value of the project's future cash flows of $1.5 million

Example, Sunk Costs

- Your company has developed a new product that is almost ready to launch. The R&D cost $0.6 million and was completed last year
- Launching requires another $1 million investment (marketing and so on), with a present value of the project's future cash flows of $1.5 million
- Proceed with the project or abort?

Example, Sunk Costs

- Your company has developed a new product that is almost ready to launch. The R&D cost $0.6 million and was completed last year
- Launching requires another $1 million investment (marketing and so on), with a present value of the project's future cash flows of $1.5 million
- Proceed with the project or abort?
  - The $0.6 million R&D is a sunk cost, the same whether you proceed or abort $\rightarrow$ Exclude it!

Example, Sunk Costs

- Your company has developed a new product that is almost ready to launch. The R&D cost \$0.6 million and was completed last year
- Launching requires another \$1 million investment (marketing and so on), with a present value of the project's future cash flows of \$1.5 million
- Proceed with the project or abort?
  - The \$0.6 million R&D is a sunk cost, the same whether you proceed or abort $\rightarrow$ Exclude it!
  - NPV $= \$1.5 - \$1 = \$0.5$ million $> 0 \rightarrow$ Proceed!

Rule c, Opportunity Costs

- A resource is almost never “free”
  - E.g., a piece of land the company already owns is not free, the firm could use it for another project, rent it out, or sell it

Rule c, Opportunity Costs

- A resource is almost never "free"
  - E.g., a piece of land the company already owns is not free, the firm could use it for another project, rent it out, or sell it
- Include the **opportunity cost** of every resource, the value of its "best alternative use"

Rule c, Opportunity Costs

- A resource is almost never "free"
  - E.g., a piece of land the company already owns is not free, the firm could use it for another project, rent it out, or sell it
- Include the **opportunity cost** of every resource, the value of its "best alternative use"
- E.g., you plan a new factory building that costs $10m to construct, on a piece of land you already own. Alternatively, you could sell the land for $3m

Rule c, Opportunity Costs

- A resource is almost never "free"
  - E.g., a piece of land the company already owns is not free, the firm could use it for another project, rent it out, or sell it
- Include the **opportunity cost** of every resource, the value of its "best alternative use"
- E.g., you plan a new factory building that costs $10m to construct, on a piece of land you already own. Alternatively, you could sell the land for $3m
- What is the actual cost of building the factory?

Rule c, Opportunity Costs

- A resource is almost never "free"
  - E.g., a piece of land the company already owns is not free, the firm could use it for another project, rent it out, or sell it
- Include the **opportunity cost** of every resource, the value of its "best alternative use"
- E.g., you plan a new factory building that costs $10m to construct, on a piece of land you already own. Alternatively, you could sell the land for $3m
- What is the actual cost of building the factory?
  - The land is not free, its opportunity cost is the $3m you give up by not selling it

Rule c, Opportunity Costs

- A resource is almost never "free"
  - E.g., a piece of land the company already owns is not free, the firm could use it for another project, rent it out, or sell it
- Include the **opportunity cost** of every resource, the value of its "best alternative use"
- E.g., you plan a new factory building that costs \$10m to construct, on a piece of land you already own. Alternatively, you could sell the land for \$3m
- What is the actual cost of building the factory?
  - The land is not free, its opportunity cost is the \$3m you give up by not selling it
  - Actual cost = \$10m + \$3m = \$13m

Rule d, Changes in NWC

- If a project needs its own NWC, the cash flows from changes in NWC belong in the analysis

Rule d, Changes in NWC

- If a project needs its own NWC, the cash flows from changes in NWC belong in the analysis
- The project's NWC often keeps changing during the project, and is often recovered at the end

Practice Problem

- A company's project runs for 3 years. In $ thousands, its revenues are 1,000 in year 1, 2,000 in year 2, and 1,000 in year 3
- The project's NWC at each date must be 10% of the following one-year period's revenues

Practice Problem

- A company's project runs for 3 years. In $ thousands, its revenues are 1,000 in year 1, 2,000 in year 2, and 1,000 in year 3
- The project's NWC at each date must be 10% of the following one-year period's revenues
- What are the cash flows from changes in NWC?

Practice Problem

- Solution ($ thousands), NWC at each date is 10% of the following one-year period's revenues,

Year	0	1	2	3
Revenues		1,000	2,000	1,000
The project's NWC	100	200	100	0
Change in NWC	+100	+100	−100	−100
Cash flow from changes in NWC	−100	−100	+100	+100

Practice Problem

- Solution ($ thousands), NWC at each date is 10% of the following one-year period's revenues,

Year	0	1	2	3
Revenues		1,000	2,000	1,000
The project's NWC	100	200	100	0
Change in NWC	+100	+100	−100	−100
Cash flow from changes in NWC	−100	−100	+100	+100

- NWC follows revenues up and down, and is fully recovered by the end of the project

Rule e, Terminal Cash Flows

- At the end, there may be cash flows from shutting down the project, the "terminal" cash flows



Rule e, Terminal Cash Flows

- At the end, there may be cash flows from shutting down the project, the "terminal" cash flows
- They can be positive
  - E.g., selling the project's fixed assets



Rule e, Terminal Cash Flows

- At the end, there may be cash flows from shutting down the project, the "terminal" cash flows
- They can be positive
  - E.g., selling the project's fixed assets
- They can also be negative
  - E.g., oil wells require "capping" when they no longer produce



Rule e, Terminal Cash Flows

- At the end, there may be cash flows from shutting down the project, the "terminal" cash flows
- They can be positive
  - E.g., selling the project's fixed assets
- They can also be negative
  - E.g., oil wells require "capping" when they no longer produce
  - But apparently many companies still do this, Bloomberg, "It Pays to Not Pay Your Debts"



Practice Problem

- A company plans a new manufacturing plant on land it already owns, but the existing buildings must be demolished. Which of the following are incremental cash flows?

a. Money already spent on engineering design of the new plant
b. The cost of a new access road put in last year
c. The market value of the land
d. The market value of the existing buildings
e. Demolition and clearance costs
f. Construction cost of the new plant
g. The initial investment in inventories of raw materials
h. Lost cash flows on other projects due to executive time spent on the new plant

Practice Problem

- A company plans a new manufacturing plant on land it already owns, but the existing buildings must be demolished. Which of the following are incremental cash flows?

a. Money already spent on engineering design of the new plant ✗
b. The cost of a new access road put in last year
c. The market value of the land
d. The market value of the existing buildings
e. Demolition and clearance costs
f. Construction cost of the new plant
g. The initial investment in inventories of raw materials
h. Lost cash flows on other projects due to executive time spent on the new plant

Practice Problem

- A company plans a new manufacturing plant on land it already owns, but the existing buildings must be demolished. Which of the following are incremental cash flows?

a. Money already spent on engineering design of the new plant ✗
b. The cost of a new access road put in last year ✗
c. The market value of the land
d. The market value of the existing buildings
e. Demolition and clearance costs
f. Construction cost of the new plant
g. The initial investment in inventories of raw materials
h. Lost cash flows on other projects due to executive time spent on the new plant

Practice Problem

- A company plans a new manufacturing plant on land it already owns, but the existing buildings must be demolished. Which of the following are incremental cash flows?

a. Money already spent on engineering design of the new plant ✗
b. The cost of a new access road put in last year ✗
c. The market value of the land ✓
d. The market value of the existing buildings
e. Demolition and clearance costs
f. Construction cost of the new plant
g. The initial investment in inventories of raw materials
h. Lost cash flows on other projects due to executive time spent on the new plant

Practice Problem

- A company plans a new manufacturing plant on land it already owns, but the existing buildings must be demolished. Which of the following are incremental cash flows?

a. Money already spent on engineering design of the new plant ✗
b. The cost of a new access road put in last year ✗
c. The market value of the land ✓
d. The market value of the existing buildings ✓
e. Demolition and clearance costs
f. Construction cost of the new plant
g. The initial investment in inventories of raw materials
h. Lost cash flows on other projects due to executive time spent on the new plant

Practice Problem

- A company plans a new manufacturing plant on land it already owns, but the existing buildings must be demolished. Which of the following are incremental cash flows?

a. Money already spent on engineering design of the new plant ✗
b. The cost of a new access road put in last year ✗
c. The market value of the land ✓
d. The market value of the existing buildings ✓
e. Demolition and clearance costs ✓
f. Construction cost of the new plant
g. The initial investment in inventories of raw materials
h. Lost cash flows on other projects due to executive time spent on the new plant

Practice Problem

- A company plans a new manufacturing plant on land it already owns, but the existing buildings must be demolished. Which of the following are incremental cash flows?

a. Money already spent on engineering design of the new plant ✗
b. The cost of a new access road put in last year ✗
c. The market value of the land ✓
d. The market value of the existing buildings ✓
e. Demolition and clearance costs ✓
f. Construction cost of the new plant ✓
g. The initial investment in inventories of raw materials
h. Lost cash flows on other projects due to executive time spent on the new plant

Practice Problem

- A company plans a new manufacturing plant on land it already owns, but the existing buildings must be demolished. Which of the following are incremental cash flows?

a. Money already spent on engineering design of the new plant ✗
b. The cost of a new access road put in last year ✗
c. The market value of the land ✓
d. The market value of the existing buildings ✓
e. Demolition and clearance costs ✓
f. Construction cost of the new plant ✓
g. The initial investment in inventories of raw materials ✓
h. Lost cash flows on other projects due to executive time spent on the new plant

Practice Problem

- A company plans a new manufacturing plant on land it already owns, but the existing buildings must be demolished. Which of the following are incremental cash flows?

a. Money already spent on engineering design of the new plant ✗
b. The cost of a new access road put in last year ✗
c. The market value of the land ✓
d. The market value of the existing buildings ✓
e. Demolition and clearance costs ✓
f. Construction cost of the new plant ✓
g. The initial investment in inventories of raw materials ✓
h. Lost cash flows on other projects due to executive time spent on the new plant ✓

Practice Problem

a. The expenditure on design is sunk and not incremental

Practice Problem

a. The expenditure on design is sunk and not incremental

b. The cost of the access road is sunk and not incremental

Practice Problem

a. The expenditure on design is sunk and not incremental
b. The cost of the access road is sunk and not incremental
c. The land could be sold or put to another use, an opportunity cost

Practice Problem

a. The expenditure on design is sunk and not incremental
b. The cost of the access road is sunk and not incremental
c. The land could be sold or put to another use, an opportunity cost
d. The buildings could be sold too, also an opportunity cost

Practice Problem

a. The expenditure on design is sunk and not incremental
b. The cost of the access road is sunk and not incremental
c. The land could be sold or put to another use, an opportunity cost
d. The buildings could be sold too, also an opportunity cost
e. Demolition and clearance costs are negative incremental cash flows

Practice Problem

a. The expenditure on design is sunk and not incremental
b. The cost of the access road is sunk and not incremental
c. The land could be sold or put to another use, an opportunity cost
d. The buildings could be sold too, also an opportunity cost
e. Demolition and clearance costs are negative incremental cash flows
f. Construction cost is the investment in fixed assets

Practice Problem

a. The expenditure on design is sunk and not incremental
b. The cost of the access road is sunk and not incremental
c. The land could be sold or put to another use, an opportunity cost
d. The buildings could be sold too, also an opportunity cost
e. Demolition and clearance costs are negative incremental cash flows
f. Construction cost is the investment in fixed assets
g. The investment in inventories is an increase in NWC

Practice Problem

a. The expenditure on design is sunk and not incremental
b. The cost of the access road is sunk and not incremental
c. The land could be sold or put to another use, an opportunity cost
d. The buildings could be sold too, also an opportunity cost
e. Demolition and clearance costs are negative incremental cash flows
f. Construction cost is the investment in fixed assets
g. The investment in inventories is an increase in NWC
h. Lost cash flows from other projects are negative incremental cash flows

Recap of Part 3

- Discount cash flows, not net income, and exclude cash flows from financing activities

Recap of Part 3

- Discount cash flows, not net income, and exclude cash flows from financing activities
- Consider all and only incremental cash flows, "would the cash flow still exist if the project did not exist?"

Recap of Part 3

- Discount cash flows, not net income, and exclude cash flows from financing activities
- Consider all and only incremental cash flows, "would the cash flow still exist if the project did not exist?"
- Include indirect effects, opportunity costs, changes in NWC, and terminal cash flows, exclude sunk costs

Module 7, Part 4

Scenario Analysis

Example, the Data Center Again

- Recall the data center from Part 2 ($ millions),
  - An investment in fixed assets of 1,000 that lasts 5 years, depreciated straight-line over 5 years

Example, the Data Center Again

- Recall the data center from Part 2 ($ millions),
  - An investment in fixed assets of 1,000 that lasts 5 years, depreciated straight-line over 5 years
  - It needs an increase in NWC of 100 at $t = 0$, recovered fully at $t = 5$

Example, the Data Center Again

- Recall the data center from Part 2 ($ millions),
  - An investment in fixed assets of 1,000 that lasts 5 years, depreciated straight-line over 5 years
  - It needs an increase in NWC of 100 at $t = 0$, recovered fully at $t = 5$
  - It brings in 240 a year after COGS and SG&A expenses

Example, the Data Center Again

- Recall the data center from Part 2 ($ millions),
  - An investment in fixed assets of 1,000 that lasts 5 years, depreciated straight-line over 5 years
  - It needs an increase in NWC of 100 at $t = 0$, recovered fully at $t = 5$
  - It brings in 240 a year after COGS and SG&A expenses
  - The corporate tax rate is 21%, and the discount rate is 8%

Example, the Data Center Again

- Recall the data center from Part 2 ($ millions),
  - An investment in fixed assets of 1,000 that lasts 5 years, depreciated straight-line over 5 years
  - It needs an increase in NWC of 100 at $t = 0$, recovered fully at $t = 5$
  - It brings in 240 a year after COGS and SG&A expenses
  - The corporate tax rate is 21%, and the discount rate is 8%
- This time, build the whole table in one go

Example, the Data Center Again

- Solution ($ millions), Components 1 and 2, then 3, then 4, then the totals

Year	0	1	2	3	4	5

Example, the Data Center Again

- Solution ($ millions), Components 1 and 2, then 3, then 4, then the totals

Year	0	1	2	3	4	5
Revenues − COGS − SG&A expenses		240	240	240	240	240

Example, the Data Center Again

- Solution ($ millions), Components 1 and 2, then 3, then 4, then the totals

Year	0	1	2	3	4	5
Revenues − COGS − SG&A expenses		240	240	240	240	240
− Depreciation		200	200	200	200	200

Example, the Data Center Again

- Solution ($ millions), Components 1 and 2, then 3, then 4, then the totals

Year	0	1	2	3	4	5
Revenues − COGS − SG&A expenses		240	240	240	240	240
− Depreciation		200	200	200	200	200
Project EBIT		**40**	**40**	**40**	**40**	**40**

Example, the Data Center Again

- Solution ($ millions), Components 1 and 2, then 3, then 4, then the totals

Year	0	1	2	3	4	5
Revenues − COGS − SG&A expenses		240	240	240	240	240
− Depreciation		200	200	200	200	200
Project EBIT		**40**	**40**	**40**	**40**	**40**
− Taxes		8.4	8.4	8.4	8.4	8.4

Example, the Data Center Again

- Solution ($ millions), Components 1 and 2, then 3, then 4, then the totals

Year	0	1	2	3	4	5
Revenues − COGS − SG&A expenses		240	240	240	240	240
− Depreciation		200	200	200	200	200
Project EBIT		**40**	**40**	**40**	**40**	**40**
− Taxes		8.4	8.4	8.4	8.4	8.4
Project net income		**31.6**	**31.6**	**31.6**	**31.6**	**31.6**

Example, the Data Center Again

- Solution ($ millions), Components 1 and 2, then 3, then 4, then the totals

Year	0	1	2	3	4	5
Revenues − COGS − SG&A expenses		240	240	240	240	240
− Depreciation		200	200	200	200	200
Project EBIT		**40**	**40**	**40**	**40**	**40**
− Taxes		8.4	8.4	8.4	8.4	8.4
Project net income		**31.6**	**31.6**	**31.6**	**31.6**	**31.6**
+ Depreciation		200	200	200	200	200

Example, the Data Center Again

- Solution ($ millions), Components 1 and 2, then 3, then 4, then the totals

Year	0	1	2	3	4	5
Revenues − COGS − SG&A expenses		240	240	240	240	240
− Depreciation		200	200	200	200	200
Project EBIT		**40**	**40**	**40**	**40**	**40**
− Taxes		8.4	8.4	8.4	8.4	8.4
Project net income		**31.6**	**31.6**	**31.6**	**31.6**	**31.6**
+ Depreciation		200	200	200	200	200
Project net income + Depreciation		**231.6**	**231.6**	**231.6**	**231.6**	**231.6**

Example, the Data Center Again

- Solution ($ millions), Components 1 and 2, then 3, then 4, then the totals

Year	0	1	2	3	4	5
Revenues − COGS − SG&A expenses		240	240	240	240	240
− Depreciation		200	200	200	200	200
Project EBIT		**40**	**40**	**40**	**40**	**40**
− Taxes		8.4	8.4	8.4	8.4	8.4
Project net income		**31.6**	**31.6**	**31.6**	**31.6**	**31.6**
+ Depreciation		200	200	200	200	200
Project net income + Depreciation		**231.6**	**231.6**	**231.6**	**231.6**	**231.6**
Cash flow from changes in NWC	−100	0	0	0	0	+100

Example, the Data Center Again

- Solution ($ millions), Components 1 and 2, then 3, then 4, then the totals

Year	0	1	2	3	4	5
Revenues − COGS − SG&A expenses		240	240	240	240	240
− Depreciation		200	200	200	200	200
Project EBIT		**40**	**40**	**40**	**40**	**40**
− Taxes		8.4	8.4	8.4	8.4	8.4
Project net income		**31.6**	**31.6**	**31.6**	**31.6**	**31.6**
+ Depreciation		200	200	200	200	200
Project net income + Depreciation		**231.6**	**231.6**	**231.6**	**231.6**	**231.6**
Cash flow from changes in NWC	−100	0	0	0	0	+100
Cash flow from capital investments	−1,000	0	0	0	0	0

Example, the Data Center Again

- Solution ($ millions), Components 1 and 2, then 3, then 4, then the totals

Year	0	1	2	3	4	5
Revenues − COGS − SG&A expenses		240	240	240	240	240
− Depreciation		200	200	200	200	200
Project EBIT		**40**	**40**	**40**	**40**	**40**
− Taxes		8.4	8.4	8.4	8.4	8.4
Project net income		**31.6**	**31.6**	**31.6**	**31.6**	**31.6**
+ Depreciation		200	200	200	200	200
Project net income + Depreciation		**231.6**	**231.6**	**231.6**	**231.6**	**231.6**
Cash flow from changes in NWC	−100	0	0	0	0	+100
Cash flow from capital investments	−1,000	0	0	0	0	0
Total cash flows	**−1,100**	**231.6**	**231.6**	**231.6**	**231.6**	**331.6**

Example, the Data Center Again

- Solution ($ millions), Components 1 and 2, then 3, then 4, then the totals

Year	0	1	2	3	4	5
Revenues − COGS − SG&A expenses		240	240	240	240	240
− Depreciation		200	200	200	200	200
Project EBIT		**40**	**40**	**40**	**40**	**40**
− Taxes		8.4	8.4	8.4	8.4	8.4
Project net income		**31.6**	**31.6**	**31.6**	**31.6**	**31.6**
+ Depreciation		200	200	200	200	200
Project net income + Depreciation		**231.6**	**231.6**	**231.6**	**231.6**	**231.6**
Cash flow from changes in NWC	−100	0	0	0	0	+100
Cash flow from capital investments	−1,000	0	0	0	0	0
Total cash flows	**−1,100**	**231.6**	**231.6**	**231.6**	**231.6**	**331.6**
NPV	**−107.23**					

- NPV = −$107.23 million, reject the project

Scenario Analysis

- The inputs behind a project's cash flows are forecasts, and forecasts can be wrong

Scenario Analysis

- The inputs behind a project's cash flows are forecasts, and forecasts can be wrong
- **Scenario analysis**, recompute the NPV under different assumptions about the inputs

Scenario Analysis

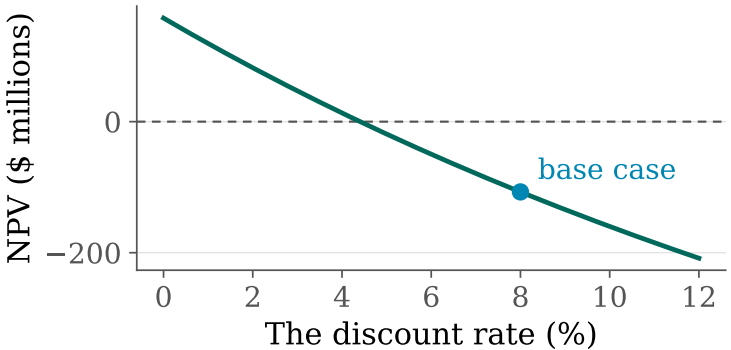
- The inputs behind a project's cash flows are forecasts, and forecasts can be wrong
- **Scenario analysis**, recompute the NPV under different assumptions about the inputs
- E.g., for the data center, how does the NPV change with the discount rate, with what it brings in a year, or with the investment in fixed assets?

Example, Scenario Analysis

- How does the NPV change with the discount rate?

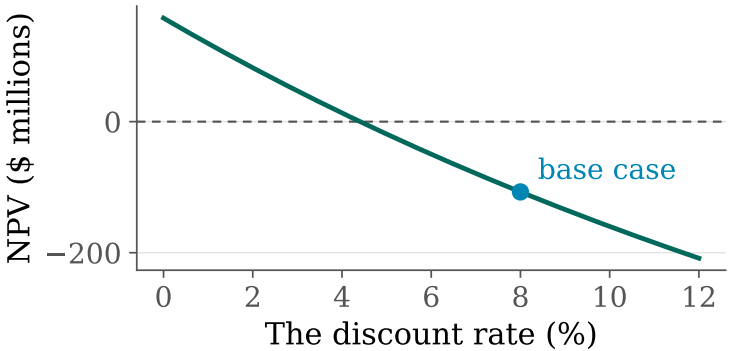
Example, Scenario Analysis

- How does the NPV change with the discount rate?



Example, Scenario Analysis

- How does the NPV change with the discount rate?



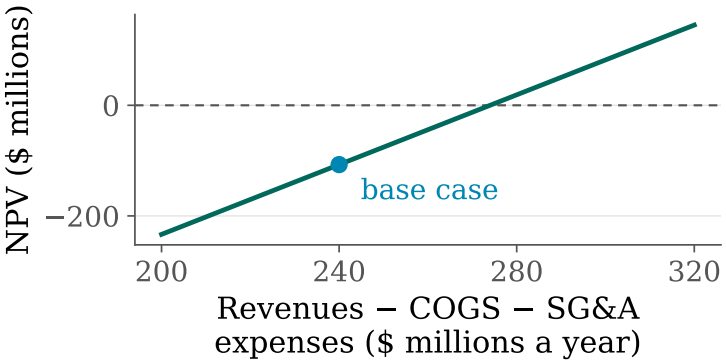
- NPV falls as the discount rate rises, positive only below the IRR of 4.41%

Example, Scenario Analysis

- How does the NPV change with what the data center brings in a year?

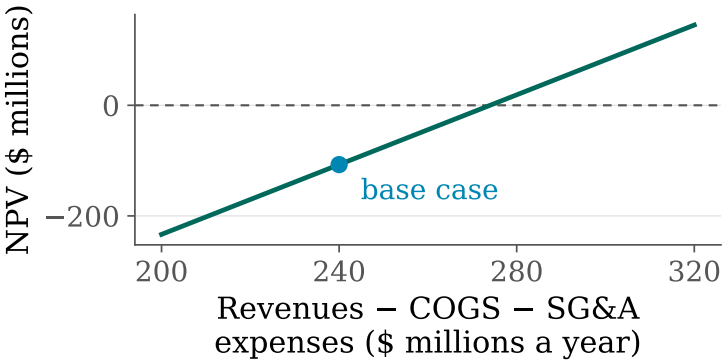
Example, Scenario Analysis

- How does the NPV change with what the data center brings in a year?



Example, Scenario Analysis

- How does the NPV change with what the data center brings in a year?



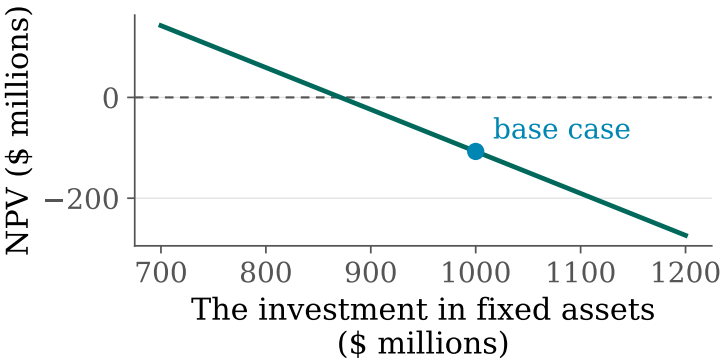
- The data center must bring in about 274 a year after COGS and SG&A expenses for the NPV to turn positive

Example, Scenario Analysis

- How does the NPV change with the investment in fixed assets?

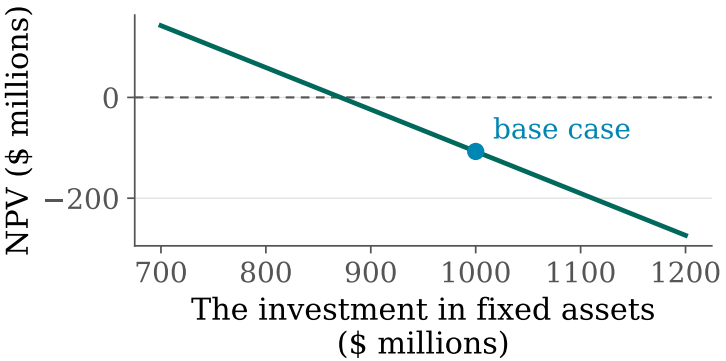
Example, Scenario Analysis

- How does the NPV change with the investment in fixed assets?



Example, Scenario Analysis

- How does the NPV change with the investment in fixed assets?



- If the data center keeps bringing in 240 a year after COGS and SG&A expenses, the investment in fixed assets must come in under about 871 for the NPV to turn positive

Using AI for Scenario Analysis

- Scenario analysis recomputes the same NPV many times, a perfect job for AI

Using AI for Scenario Analysis

- Scenario analysis recomputes the same NPV many times, a perfect job for AI
- **Never ask the AI to directly give you the answer**
  - It can make up or miscompute numbers, and you cannot check where they came from

Using AI for Scenario Analysis

- Scenario analysis recomputes the same NPV many times, a perfect job for AI
- **Never ask the AI to directly give you the answer**
  - It can make up or miscompute numbers, and you cannot check where they came from
- Ask the AI for **code or a spreadsheet that produces the answer**
  - You can read it, check it, and rerun it with new inputs

Using AI for Scenario Analysis

- Scenario analysis recomputes the same NPV many times, a perfect job for AI
- **Never ask the AI to directly give you the answer**
  - It can make up or miscompute numbers, and you cannot check where they came from
- Ask the AI for **code or a spreadsheet that produces the answer**
  - You can read it, check it, and rerun it with new inputs
- In-class demo, I give the AI a prompt and run the code it returns

Module 7, Part 5

Practice Problem

Practice Problem

- A company makes AI chips and just developed a new AI chip design
- The new design needs an investment in fixed assets (machinery) of $6 million, depreciated straight-line over 6 years

Practice Problem

- A company makes AI chips and just developed a new AI chip design
- The new design needs an investment in fixed assets (machinery) of \$6 million, depreciated straight-line over 6 years
- It needs an increase in NWC of \$250,000 at $t = 0$, recovered fully at $t = 6$

Practice Problem

- A company makes AI chips and just developed a new AI chip design
- The new design needs an investment in fixed assets (machinery) of \$6 million, depreciated straight-line over 6 years
- It needs an increase in NWC of \$250,000 at $t = 0$, recovered fully at $t = 6$
- It sells 1,000 AI chips in year 1, growing at 5% a year over the six years. Each AI chip sells for \$4,000, and its COGS is \$2,000. There are no SG&A expenses

Practice Problem

- A company makes AI chips and just developed a new AI chip design
- The new design needs an investment in fixed assets (machinery) of \$6 million, depreciated straight-line over 6 years
- It needs an increase in NWC of \$250,000 at $t = 0$, recovered fully at $t = 6$
- It sells 1,000 AI chips in year 1, growing at 5% a year over the six years. Each AI chip sells for \$4,000, and its COGS is \$2,000. There are no SG&A expenses
- The project ends after 6 years. The corporate tax rate is 21%, and the discount rate is 12%

Practice Problem

- A company makes AI chips and just developed a new AI chip design
- The new design needs an investment in fixed assets (machinery) of \$6 million, depreciated straight-line over 6 years
- It needs an increase in NWC of \$250,000 at $t = 0$, recovered fully at $t = 6$
- It sells 1,000 AI chips in year 1, growing at 5% a year over the six years. Each AI chip sells for \$4,000, and its COGS is \$2,000. There are no SG&A expenses
- The project ends after 6 years. The corporate tax rate is 21%, and the discount rate is 12%
- What is the project's NPV?

Solution, Component 1, Project Net Income

- Build the project's income statement year by year ($ thousands),

Year	0	1	2	3	4	5	6

Solution, Component 1, Project Net Income

- Build the project's income statement year by year ($ thousands),

Year	0	1	2	3	4	5	6
AI chips sold		1,000	1,050	1,103	1,158	1,216	1,276

Solution, Component 1, Project Net Income

- Build the project's income statement year by year ($ thousands),

Year	0	1	2	3	4	5	6
AI chips sold		1,000	1,050	1,103	1,158	1,216	1,276
Revenues		4,000	4,200	4,410	4,631	4,862	5,105

- Revenues = \$4,000× AI chips sold, which grow 5% a year, so \$4 million in year 1

Solution, Component 1, Project Net Income

- Build the project's income statement year by year ($ thousands),

Year	0	1	2	3	4	5	6
AI chips sold		1,000	1,050	1,103	1,158	1,216	1,276
Revenues		4,000	4,200	4,410	4,631	4,862	5,105
− COGS		2,000	2,100	2,205	2,315	2,431	2,553

- Revenues = \$4,000× AI chips sold, which grow 5% a year, so \$4 million in year 1

Solution, Component 1, Project Net Income

- Build the project's income statement year by year ($ thousands),

Year	0	1	2	3	4	5	6
AI chips sold		1,000	1,050	1,103	1,158	1,216	1,276
Revenues		4,000	4,200	4,410	4,631	4,862	5,105
− COGS		2,000	2,100	2,205	2,315	2,431	2,553
− Depreciation		1,000	1,000	1,000	1,000	1,000	1,000

- Revenues = \$4,000× AI chips sold, which grow 5% a year, so \$4 million in year 1
- Depreciation = 6,000/6 = 1,000 a year

Solution, Component 1, Project Net Income

- Build the project's income statement year by year ($ thousands),

Year	0	1	2	3	4	5	6
AI chips sold		1,000	1,050	1,103	1,158	1,216	1,276
Revenues		4,000	4,200	4,410	4,631	4,862	5,105
− COGS		2,000	2,100	2,205	2,315	2,431	2,553
− Depreciation		1,000	1,000	1,000	1,000	1,000	1,000
Project EBIT		**1,000**	**1,100**	**1,205**	**1,315**	**1,431**	**1,553**

- Revenues = \$4,000× AI chips sold, which grow 5% a year, so \$4 million in year 1
- Depreciation = 6,000/6 = 1,000 a year

Solution, Component 1, Project Net Income

- Build the project's income statement year by year ($ thousands),

Year	0	1	2	3	4	5	6
AI chips sold		1,000	1,050	1,103	1,158	1,216	1,276
Revenues		4,000	4,200	4,410	4,631	4,862	5,105
− COGS		2,000	2,100	2,205	2,315	2,431	2,553
− Depreciation		1,000	1,000	1,000	1,000	1,000	1,000
Project EBIT		**1,000**	**1,100**	**1,205**	**1,315**	**1,431**	**1,553**
− Taxes		210	231	253	276	301	326

- Revenues = \$4,000× AI chips sold, which grow 5% a year, so \$4 million in year 1
- Depreciation = 6,000/6 = 1,000 a year

Solution, Component 1, Project Net Income

- Build the project's income statement year by year ($ thousands),

Year	0	1	2	3	4	5	6
AI chips sold		1,000	1,050	1,103	1,158	1,216	1,276
Revenues		4,000	4,200	4,410	4,631	4,862	5,105
− COGS		2,000	2,100	2,205	2,315	2,431	2,553
− Depreciation		1,000	1,000	1,000	1,000	1,000	1,000
Project EBIT		**1,000**	**1,100**	**1,205**	**1,315**	**1,431**	**1,553**
− Taxes		210	231	253	276	301	326
Project net income		**790**	**869**	**952**	**1,039**	**1,130**	**1,227**

- Revenues = \$4,000× AI chips sold, which grow 5% a year, so \$4 million in year 1
- Depreciation = 6,000/6 = 1,000 a year

Solution, Component 2, Depreciation

- Depreciation is not a cash flow, add it back ($ thousands),

Year	0	1	2	3	4	5	6

Solution, Component 2, Depreciation

- Depreciation is not a cash flow, add it back ($ thousands),

Year	0	1	2	3	4	5	6
Project net income		790	869	952	1,039	1,130	1,227

Solution, Component 2, Depreciation

- Depreciation is not a cash flow, add it back ($ thousands),

Year	0	1	2	3	4	5	6
Project net income		790	869	952	1,039	1,130	1,227
+ Depreciation		1,000	1,000	1,000	1,000	1,000	1,000

Solution, Component 2, Depreciation

- Depreciation is not a cash flow, add it back ($ thousands),

Year	0	1	2	3	4	5	6
Project net income		790	869	952	1,039	1,130	1,227
+ Depreciation		1,000	1,000	1,000	1,000	1,000	1,000
Project net income + Depreciation		**1,790**	**1,869**	**1,952**	**2,039**	**2,130**	**2,227**

- The cash for the machinery went out at $t = 0$, it is counted in Component 4

Solution, Component 3, Cash Flow from Changes in NWC

- The project's NWC is 250 while it operates, recovered at $t = 6$ ($ thousands),

Year	0	1	2	3	4	5	6

Solution, Component 3, Cash Flow from Changes in NWC

- The project's NWC is 250 while it operates, recovered at $t = 6$ ($ thousands),

Year	0	1	2	3	4	5	6
The project's NWC	250	250	250	250	250	250	0

Solution, Component 3, Cash Flow from Changes in NWC

- The project's NWC is 250 while it operates, recovered at $t = 6$ ($ thousands),

Year	0	1	2	3	4	5	6
The project's NWC	250	250	250	250	250	250	0
Change in NWC	+250	0	0	0	0	0	−250

Solution, Component 3, Cash Flow from Changes in NWC

- The project's NWC is 250 while it operates, recovered at $t = 6$ ($ thousands),

Year	0	1	2	3	4	5	6
The project's NWC	250	250	250	250	250	250	0
Change in NWC	+250	0	0	0	0	0	−250
Cash flow from changes in NWC	**−250**	**0**	**0**	**0**	**0**	**0**	**+250**

- An increase in NWC is a negative cash flow, and a decrease is a positive cash flow

Solution, Component 4, Cash Flow from Capital Investments

- Investment in fixed assets (the machinery) at $t = 0$ ($ thousands),

Year	0	1	2	3	4	5	6

Solution, Component 4, Cash Flow from Capital Investments

- Investment in fixed assets (the machinery) at $t = 0$ ($ thousands),

Year	0	1	2	3	4	5	6
Investment in fixed assets	−6,000	0	0	0	0	0	0

Solution, Component 4, Cash Flow from Capital Investments

- Investment in fixed assets (the machinery) at $t = 0$ ($ thousands),

Year	0	1	2	3	4	5	6
Investment in fixed assets	−6,000	0	0	0	0	0	0
Cash flow from capital investments	**−6,000**	**0**	**0**	**0**	**0**	**0**	**0**

Solution, Total Cash Flows

- Add up the four components year by year ($ thousands),

Year	0	1	2	3	4	5	6

Solution, Total Cash Flows

- Add up the four components year by year ($ thousands),

Year	0	1	2	3	4	5	6
Project net income + Depreciation	0	1,790	1,869	1,952	2,039	2,130	2,227

Solution, Total Cash Flows

- Add up the four components year by year ($ thousands),

Year	0	1	2	3	4	5	6
Project net income + Depreciation	0	1,790	1,869	1,952	2,039	2,130	2,227
Cash flow from changes in NWC	−250	0	0	0	0	0	+250

Solution, Total Cash Flows

- Add up the four components year by year ($ thousands),

Year	0	1	2	3	4	5	6
Project net income + Depreciation	0	1,790	1,869	1,952	2,039	2,130	2,227
Cash flow from changes in NWC	−250	0	0	0	0	0	+250
Cash flow from capital investments	−6,000	0	0	0	0	0	0

Solution, Total Cash Flows

- Add up the four components year by year ($ thousands),

Year	0	1	2	3	4	5	6
Project net income + Depreciation	0	1,790	1,869	1,952	2,039	2,130	2,227
Cash flow from changes in NWC	−250	0	0	0	0	0	+250
Cash flow from capital investments	−6,000	0	0	0	0	0	0
Total cash flows	**−6,250**	**1,790**	**1,869**	**1,952**	**2,039**	**2,130**	**2,477**

- At $t = 0$, $-6{,}000 - 250 = -6{,}250$. At $t = 6$, $2{,}227 + 250 = 2{,}477$

Solution, NPV and the Decision

- Discount the total cash flows at the 12% discount rate ($ thousands),

Solution, NPV and the Decision

- Discount the total cash flows at the 12% discount rate ($ thousands),

$$\text{NPV} = -6{,}250 + \frac{1{,}790}{1.12} + \frac{1{,}869}{1.12^2} + \frac{1{,}952}{1.12^3} + \cdots + \frac{2{,}477}{1.12^6} = 1{,}987$$

Solution, NPV and the Decision

- Discount the total cash flows at the 12% discount rate ($ thousands),

$$\text{NPV} = -6{,}250 + \frac{1{,}790}{1.12} + \frac{1{,}869}{1.12^2} + \frac{1{,}952}{1.12^3} + \cdots + \frac{2{,}477}{1.12^6} = 1{,}987$$

- NPV > 0, so the company should launch the new design

Solution, NPV with the Cash Flow Worksheet

- The same NPV on the TI BA II Plus cash flow worksheet ($ thousands)

Solution, NPV with the Cash Flow Worksheet

- The same NPV on the TI BA II Plus cash flow worksheet ($ thousands)
- Every total cash flow differs, so enter each one once and leave its frequency at 1

Enter	Keys
	CF, **2nd**, **CLR Work**
−6250	**ENTER**, ↓
1790	**ENTER**, ↓ ↓
1869	**ENTER**, ↓ ↓
1952	**ENTER**, ↓ ↓
2039	**ENTER**, ↓ ↓
2130	**ENTER**, ↓ ↓
2477	**ENTER**
	NPV (the calculator asks for I)
12	**ENTER**, ↓
	CPT
1,986.93	

Solution, NPV with the Cash Flow Worksheet

- The same NPV on the TI BA II Plus cash flow worksheet ($ thousands)
- Every total cash flow differs, so enter each one once and leave its frequency at 1
- NPV $= 1{,}986.93 \approx 1{,}987 > 0$, the same answer, launch the new design

Enter	Keys
	CF, 2nd, CLR Work
−6250	ENTER, ↓
1790	ENTER, ↓ ↓
1869	ENTER, ↓ ↓
1952	ENTER, ↓ ↓
2039	ENTER, ↓ ↓
2130	ENTER, ↓ ↓
2477	ENTER
	NPV (the calculator asks for I)
12	ENTER, ↓
	CPT
1,986.93	

Key Takeaways

- Cash flow from a project adds project net income, depreciation, cash flow from changes in NWC, and cash flow from capital investments

Key Takeaways

- Cash flow from a project adds project net income, depreciation, cash flow from changes in NWC, and cash flow from capital investments
- Discount cash flows, not net income, and exclude cash flows from financing activities

Key Takeaways

- Cash flow from a project adds project net income, depreciation, cash flow from changes in NWC, and cash flow from capital investments
- Discount cash flows, not net income, and exclude cash flows from financing activities
- Only incremental cash flows count, include indirect effects, opportunity costs, changes in NWC, and terminal cash flows, exclude sunk costs

Key Takeaways

- Cash flow from a project adds project net income, depreciation, cash flow from changes in NWC, and cash flow from capital investments
- Discount cash flows, not net income, and exclude cash flows from financing activities
- Only incremental cash flows count, include indirect effects, opportunity costs, changes in NWC, and terminal cash flows, exclude sunk costs
- Lay the years out in a table, total the cash flows, discount, and apply the NPV rule

Thank you!

Please let me know if you have any questions.